

Hardik Jindal

Third Year Undergraduate
Department of Electrical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2022-Present	B.Tech	Indian Institute of Technology Kanpur	8.6/10
2022	CBSE (XII)	Bharti Public School, Delhi	94%
2020	CBSE (X)	D.L.F Public School, Ghaziabad	96%

Scholastic Achievements

- Secured **All India Rank 889** in **JEE Advanced 2022** among the 2.5 Lakh shortlisted candidates (2022)
- Secured **All India Rank 2257** in **JEE Mains 2022** amongst over 1.5 Million candidates (2022)
- Secured **All India Rank 894** in **KVPY SX** conducted by IISc, Bangalore from over 1.5 Lakh candidates (2022)
- Recipient of the **Dr. S.L. Batra Excellence Award** for exemplary academic performance in the academic year 2021-22 (2022)
- Achieved **2nd position worldwide** in IROS Simulation Humanoid Robot Wrestling Competition organized by **IEEE** (2023)
- Secured **1st Position** in **SnT Hackathon 2023** conducted by **Science and Technology (SnT) Council**, IIT Kanpur (2023)
- Qualified for **2nd Round** of **AMEX Campus Challenge - Decision Science Track!**, organized by **American Express** (2024)

Incoming Internships

- Incoming Member of Technical Staff Summer'25 Intern at **Oracle India**

Work Experience

Google Summer of Code'24 | Ceylon Computer Science Institute | Mentors: N.Weerasekara, T.Cui |  |  (May'24 - Aug'24)

- Developed a **RAG** based **RESTful** API in **Flask** for retrieval of cybersecurity news from **100+** articles by leading news outlets
- Transitioned the codebase to open LLMs like **LLaMA3**, **Gemma**, **MistralAI** by integrating a **HuggingFace-based** endpoint
- Integrated **Langchain** to establish a **LLM chain** so as to facilitate structured data processing and formatting for news handling
- Utilized **Pinecone** as a **vector database** in order to efficiently store and significantly reduce retrieval times for **data embeddings**
- Developed unit tests using **unittest** for Flask app, validating API functionality, HTTP status code, and response payload integrity

mSense AI  | Software Development Intern (Sep'23 - Oct'23)

- Experimented with **PaddleOCR** and **EasyOCR** frameworks to create customized **text detection** and **recognition** models
- Conceptualized and integrated a custom **training pipeline** for **text detection** and **text recognition** with **MKL-DNN** framework
- Adjusted parameters such as **dilation**, **angle classifier**, **batch size**, and **CPU threads** to determine the most optimum techniques

Major Competitions

IEEE IROS Humanoid Robot Wrestling Simulation Competition | Team Humanoid, IITK |  (Aug'23 - Oct'23)

Objective	• To simulate a calibrated NAO robot and devise a strategy for virtual wrestling matches in the WeBots environment
Approach	• Integrated a YOLOv5 model for recognizing the position , depth and spatial orientation of opponent in real-time • Implemented image segmentation and contour detection techniques, including Hough transform for detection
Result	• Achieved 2nd position worldwide , competing against 30+ international teams from various universities

DevRev AI Agent 007: Tooling up for Success | Inter IIT Tech Meet 12.0 | High Prep Event |  (Oct'23 - Dec'23)

Objective	• To develop a Dynamic Tool Orchestration system for LLMs to identify the appropriate set of tools for a query
Approach	• Finetuned the LLaMA2 model using techniques like QLoRA to generate accurate and context-aware responses • Devised a custom Retriever using FAISS along with L2 Metric to retrieve tools & examples relevant to the query • Utilized OpenAI's SOTA text-embedding-ada-002 as an embedding model and stanza for efficient clause parsing
Result	• Achieved an EM score of 64% , optimizing average inference cost to USD 0.007 with GPT 4 and 3.5 Turbo LLM

Key Projects

AI-Based Cost Optimization in Indian Railways | Prof. Gururaj Mirle Vishwanath, IIT Kanpur (Aug'23 - Oct'23)

- Used **ensemble** modeling with **XGBoost** and **AdaBoost** to forecast railway loads, optimizing costs and power consumption
- Applied **feature engineering** techniques along with **cross-validation** and **hyperparameter** tuning to improve model efficiency

Scam Detection | Uttar Pradesh Police (Jun'24 - Present)

- Utilized **Tesseract OCR** and **Easy OCR** for **text detection** and **recognition** from screenshots across various messaging platforms
- Developed and fine-tuned a **BERT** based text classification model to detect scam messages from OCR extracted text

NFS Goes Online | Project Mentor | Programming Club, IITK |  (May'24 - Jul'24)

- Mentored **15 mentees** to understand the fundamentals of the **Rust** Programming language and various **network protocols**
- Demonstrated Rust's **concurrency features** and **ownership model** for building efficient, secure network applications
- Developed a single threaded **web-server** in Rust, capable of listening to a **TCP connection** and **HTTP request** handling

Campus Fora | Programming Club, IITK |  (May'23 - Aug'23)

- Developed a forums based **web application** with an expected user base of **4000** students for **efficient knowledge-sharing**
- Fabricated an adaptive frontend on **NextJS** with features like **lazy loading** and search optimizations using **Redis cache**

Technical Skills

- **Programming Languages:** C | C++ | Python | Bash | Rust
- **Utilities:** Git | GitHub | Linux | HuggingFace | $\LaTeX$ | Markdown
- **Libraries:** Numpy | Pandas | Scikit-Learn | Matplotlib | Tensorflow | Transformers | OpenCV | LangChain | Ollama | NLTK

Positions of Responsibility

Coordinator | Programming Club IITK 

(Apr'24 - Present)

Leadership	- Leading a two-tier team comprising of 25 secretaries to foster a culture of programming in the campus community - Organized 100+ shortlisting interviews to recruit secretaries & managing a budget of INR 1.7 Lakh for events
Initiatives	- Collaborated with Trumio, Canonical and UP Police to work on projects spanning various domains of programming - Mentored 100+ juniors shortlisted from 500+ applications in 7 projects spanning domains of ML, WebDev, etc - Conducting contests and hackathons for the campus community in order to promote a programming culture
Impact	- Achieved 1.5x reach, garnering more than 35K impressions on Student Search & 4K registrations on Puppy Love - Created innovative tools and platforms to assist the UP Police, deployed for over 20K+ officers across 4 districts

Relevant Courses (*:Excellent Performance)

Data Structure and Algorithms Introduction to ML Control Systems and Analysis	Fundamentals of Computing* Introduction to Electronics* Digital Electronics	Probability and Statistics Linear Algebra and ODE Microelectronics	Partial Differential Equations* Signal, Systems and Network Principles of Communication
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